

ECM–TQAG: Evidence-Contracted Multimodal Generation of Textbook Question–Answer Items from Scanned Documents

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Abstract—We present ECM–TQAG, a protocol for generating evidence-grounded multimodal question–answer items from scanned documents. For each source chunk the protocol fixes a conditioning bundle of recognised text, document structure, and hash-bound figure crops; an evidence contract then requires the generator, within a single call, to commit to a verbatim text quotation, an answer planned before the question, and a figure identified by content hash, and to keep this evidence tuple fixed while realising the item. A deterministic verifier recomputes the returned tuple against the bundle and admits the item only if every component reproduces, with no repair and no human judgement. In a paired experiment over 24 figure-bearing chunks from scanned Vietnamese-law holdings, the prompt is the only quantity that varies: the contracted prompt admits 22/24 items against 16/24 and 12/24 for two control prompts, with both paired contrasts significant under multiplicity control, and it eliminates quotation from text that is rendered only inside page images.

Index Terms—multimodal question generation, document intelligence, evidence grounding, structured decoding, legal education

I. INTRODUCTION AND RELATED WORK

Scanned textbooks and institutional documents combine prose, diagrams, and tables, motivating textbook QA [1] and its generation counterpart. A fluent generated question may still assert unsupported content [2] or may only appear multimodal while remaining answerable from text alone [3], [4]. Existing document models fuse text, layout, and pixels [5]–[7] or read pixels directly [8], [9], while DocVQA and ChartQA assess answers [10], [11]; educational and visual question generation motivate the construction task [12]–[14].

Our contribution is the ECM–TQAG protocol itself: evidence support is stated as a contract on one generation call, and that contract is made checkable by a deterministic verifier

that recomputes the declared evidence against the source. The protocol is evaluated by a paired design in which each source chunk serves as its own control, so any difference in the measured outcome is attributed to the contract alone. Section II specifies the protocol, Section III instantiates the experiment, and Sections IV–V report the census.

II. METHOD

ECM–TQAG has three components: a conditioning bundle that fixes what may serve as evidence, an evidence contract that constrains how an item is produced from that bundle, and a deterministic verifier that turns each returned item into a binary outcome. Fig. 1 shows the architecture, including the data pipeline that builds the bundles.

A. Conditioning bundle

Each source chunk c is represented by

$$E_c = (T_c, L_c, \mathcal{I}_c), \quad (1)$$

where T_c is the recognised page text, L_c the document structure (headings, figure and table regions), and $\mathcal{I}_c$ the set of figure crops bound by content hash. Evidence outside E_c is inadmissible for a chunk, so the contract is checkable rather than declarative.

B. Evidence contract

One multimodal generator g is fixed for the whole protocol. Under arm $a \in \mathcal{A} = \{\text{ECM}, \text{DIR}, \text{STR}\}$, one call produces one candidate item,

$$y_{c,a} = g(\pi_a, E_c), \quad (2)$$

where π_a is the arm prompt and the decoding settings, the question type t_c , and the response schema are identical across arms. The ECM contract adds four operations to that shared instruction: it selects a verbatim quotation q from T_c together

with a figure h from $\mathcal{I}_c$; it plans the answer α before wording the question; it holds the evidence tuple

$$\tau = (t, q, \alpha, h) \quad (3)$$

fixed while the item is realised; and it self-checks schema conformance. All four operations are instructions within the single call of (2); because the provider returns only the final structured item, intermediate reasoning never enters the evidence record. The two control arms share the same call structure; the direct arm simply requests one multimodal item, and the structured no-contract arm additionally emphasises exact schema completion.

C. Deterministic verifier

Verification recomputes $\tau(y)$ against E_c . The gate is the conjunction

$$G(y, E_c) = \mathbb{K}\{t = t_c\} \cdot \mathbb{K}\{q \sqsubseteq T_c\} \cdot \mathbb{K}\{h \in \mathcal{I}_c\} \cdot \mathbb{K}\{\sigma(y)\} \cdot \mathbb{K}\{\mu(y)\}, \quad (4)$$

where $q \sqsubseteq T_c$ requires the quotation to be a contiguous substring of the recognised text, σ is exact schema conformance with nonempty question, answer, visual description, and rationale, and μ requires, for multiple choice, four distinct options whose marked option equals α . Because G is evaluated on the response as returned, admission is a property of the response itself. The mechanical admission of chunk c under arm a is

$$A(c, a) = G(y_{c,a}, E_c), \quad (5)$$

computed entirely by the verifier. Passing G certifies source consistency only; it does not establish truth, legal correctness, or pedagogical value.

D. Paired identification

Because g , E_c , t_c , the schema, and the decoding settings are common to all arms, the paired difference $A(c, \text{ECM}) - A(c, a')$ varies only with the prompt. Each chunk therefore acts as its own control, and the contract is tested by exact paired comparisons with prespecified multiplicity control. In addition, two arm-blind model raters from independent families supply secondary semantic scores; these are treated as exploratory in this work.

III. EXPERIMENTAL SETUP

a) Material.: The frame is a bounded census of 24 figure-bearing chunks, not a population sample. The chunks come from scanned holdings of the University of Law, Hue University: reference monographs, conference proceedings, course textbooks, student research reports, undergraduate theses, and institutional process and form documents, used for internal research only; no page image is redistributed. The frame retains chunks whose figure is present, whose crop is resolvable, and whose text is long enough to support a verbatim quotation (Fig. 1).

TABLE I
MECHANICAL ADMISSION A OVER THE 24-CHUNK FRAME. AN ATTEMPT IS ADMITTED ONLY IF ITS RETURNED EVIDENCE TUPLE RECOMPUTES AGAINST THE CHUNK'S OWN BUNDLE.

Arm	Admitted	Rate	Contract violations
ECM (contracted)	22/24	0.92	2
Direct	16/24	0.67	7
Structured, no contract	12/24	0.50	9

b) Arms and models.: Each chunk receives one call under each of the three arms of (2). The generator is Qwen3-VL-8B-Instruct, held fixed across arms so that no model change can confound the prompt contrast. The two arm-blind raters are Claude Opus 5 and GPT-5.6 Sol, from independent model families.

c) Quantities and tests.: The measured outcome is the mechanical admission A of (5), reported over all assigned chunks: a chunk whose item fails any gate condition contributes zero to the rate. Each paired contrast compares the contracted arm with one control arm by an exact McNemar test; the two p -values are adjusted by Holm's step-down procedure at family-wise $\alpha = 0.05$ (thresholds 0.025 and 0.05).

IV. RESULTS

A. Census outcomes

The census comprises $24 \times 3 = 72$ chunk-arm attempts, of which 50 were admitted and 22 failed verification.

B. Mechanical admission

Table I reports $A(c, a)$, the share of chunks whose returned item reproduces against its own bundle. The quantity is computed entirely by the verifier, so it is reproducible from the released records.

Both paired contrasts are significant. Against the direct arm, 6 chunks are admitted only under the contract and none in the opposite direction, giving an exact McNemar $p = 0.031$. Against the structured no-contract arm there are 11 against 1, giving $p = 0.006$. Under Holm control at family $\alpha = 0.05$ the smaller p is compared with 0.025 and the larger with 0.05, so both contrasts reject. The contract therefore raises the rate at which a single call yields a source-consistent item, with each chunk as its own control.

C. Failure structure

Of the 22 failures, 18 violate the verbatim-quotation condition $q \sqsubseteq T_c$, 2 return no single decodable JSON object, and 2 return a malformed field set. The verbatim violations fall into three kinds.

In ten attempts the quotation is text rendered *inside the page image* and absent from the recognised text: independent recognition of each figure crop matches eight of the quotations to the image text completely and two in substantial part. This is the failure mode the verbatim rule is designed to prevent; it occurs 4 times in the direct arm and 6 times in the structured no-contract arm, and never in the contracted arm.

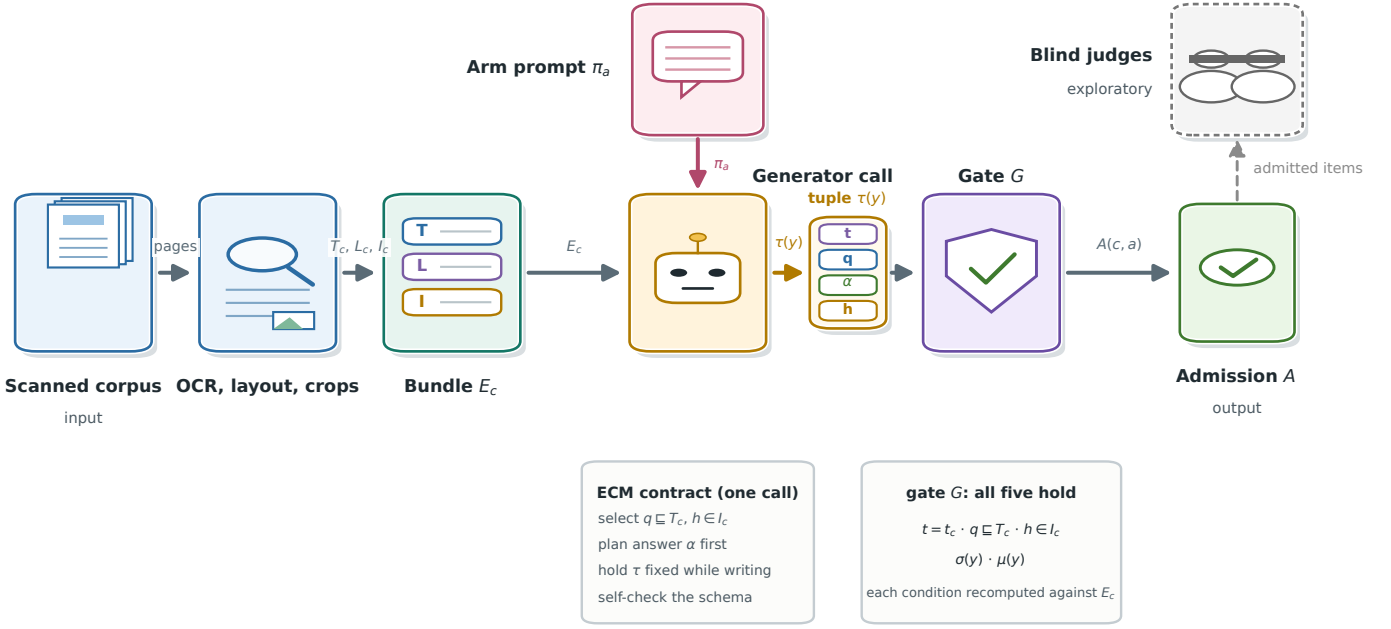


Fig. 1. ECM-TQAG architecture and data pipeline. Left: scanned holdings are rendered page by page; recognition, layout analysis, and figure-crop extraction produce the conditioning bundle $E_c = (T_c, L_c, I_c)$. Centre: the bundle and an arm prompt π_a enter one generation call; under the ECM arm the call carries the four contract operations listed in the panel. Right: the returned evidence tuple $\tau(y) = (t, q, \alpha, h)$ is recomputed by the deterministic gate G , whose five conditions are evaluated against the same bundle without repair, yielding the binary admission $A(c, a)$ used for all reported contrasts. Judging of admitted items by two arm-blind raters is a secondary, exploratory measurement.

Four further attempts differ from the source only in the capitalisation of the first character; the gate normalises Unicode composition and whitespace but not letter case, so they are rejected. Admitting them post hoc would give 23/24, 17/24, and 14/24 with $p = 0.031$ and $p = 0.012$, leaving the direction, the ordering, and both rejections unchanged.

The remaining four quotations are composed or paraphrased rather than copied, spread across the three arms.

V. DISCUSSION

a) Effect of the contract.: The contract changes the rate at which one call produces an item whose declared evidence actually recomputes against the source: 0.92 against 0.67 and 0.50, with both paired contrasts significant after Holm adjustment. The mechanism is visible in the failures rather than inferred from the totals: the dominant failure of the uncontracted arms is a quotation copied from lettering inside the page image, which no textual source supports, and it occurs ten times without the contract and never with it. Because admission is computed by the verifier alone, the result requires no human rating and can be recomputed from the released records.

b) Limitations.: The frame is a bounded census from one institution’s holdings, and the two rejections concern mechanical admission on that frame rather than any population claim. The generator is a single open-weight multimodal model, and admission certifies source consistency, not truth, legal correctness, or pedagogical value. The secondary semantic

ratings are exploratory and are not calibrated against blinded human ratings.

c) Future work.: Three directions would raise the protocol’s reach. First, visual necessity can be measured by an ablation arm that answers from text alone, turning a rated score into a direct measurement. Second, the judged criteria can be calibrated against blinded human ratings before they are used as endpoints. Third, the protocol can be extended to corpora whose figures carry information absent from the prose and to other generators, with any refinement of the verbatim rule fixed before new calls are issued.

VI. CONCLUSION

ECM-TQAG states evidence support as a contract on one generation call and makes that contract checkable by recomputing the declared evidence tuple against the source bundle. In a paired census of 24 figure-bearing chunks in which the prompt is the only quantity that varies, the contract significantly raises the rate at which a single call yields a source-consistent item, and it eliminates the dominant failure mode of the control arms, namely quotation from text rendered only inside page images. Admission certifies source consistency rather than truth or pedagogical value, and extending the protocol to semantic validity at scale remains open.

DATA AVAILABILITY

The evidence consists of the frozen protocol, the evaluation instrument, and the complete census records with their primary analysis. Source material is held by the University of

Law, Hue University, for internal research only: copyrighted page images, figure crops, and raw model responses are not distributed.

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